Minority Salience and Crime

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Abstract

This paper investigates how changes in the salience of Muslim communities affect local crime rates. Leveraging police recorded crime data, I use a Bartik-style estimation strategy exploiting the exogenous within-neighbourhood variation in salience generated by Islamic terrorist attacks. The results show a large increase in violent crime ($\approx 3.2\%$), which is particularly pronounced for domestic attacks ($\approx 5\%$). Heterogeneity analysis reveals that areas with lower education, religious diversity and migrant inflows experience more substantial violent backlash. Similarly, a wide gap is found between areas with high vs. low electoral support for right-wing and anti-immigrant parties. Findings are corroborated using a matched difference-in-difference approach and a battery of robustness checks. (*JEL* D91, J15, K42, Z12, Z13)

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1 Introduction

The last two decades have seen substantial growth in ethnic and religious diversity across many Western economies.[1] These major societal changes have come with heterogeneous effects and coincided with strict policies of fiscal consolidation following the 2008/09 Financial Crisis. The subsequent growth of right-wing populism, expressing opposition to migrants and other minorities, has prompted intense debate about whether this emerging sentiment has economic or socio-cultural roots, and what governments can do to instead promote trust and social cohesion (Alesina and Tabellini, 2024; Guriev and Papaioannou, 2022).

In this paper I contribute to the debate, by exploring a less documented but more severe symptom of growing hostility towards minorities: violent backlash. In particular, I investigate the relationship between changes in the salience of Muslim communities and neighbourhood-level crime, and evaluate how important local differences in socio-economic characteristics and political preferences are in explaining the magnitude of these often violent responses. To this end, I use the example of the UK which not only has a large Muslim population, but has enacted significant austerity policies and left the European Union partly on the promise of greater control over its borders (Becker and Fetzer, 2017; Fetzer, 2019). [2]

Crucially, changes in the salience of minority groups, whereby the attention of the public is disproportionally drawn to concerns about these communities, can magnify existing prejudices and expose their sources, whether economic or cultural (Bonomi et al., 2021; Bordalo et al., 2013). For instance, during Ramadan, electoral support for far-right or nationalist parties in Germany increases in areas closer to Mosques, where locals become more aware of their Muslim neighbours (Colussi et al., 2021). However, to see how societies respond to external threats, I focus on a more extreme and negative source of shocks to salience: Islamic terrorist attacks. In a well-integrated society such events should not be attributed or blamed on local Muslims communities, and no backlash should be observed (Shayo, 2020). Sadly, this is not what I find.

[1] From 2000 to 2020 the share of foreign-born residents in OECD countries rose from 8.9% to 12.2%.

[2] Moreover, the number of hate crimes in the UK has grown year-on-year up until the start of the pandemic. For more information see: `https://www.gov.uk/government/statistics/hate-crime-england-and-wales-year-ending-march-2025/hate-crime-england-and-wales-year-ending-march-2025`.

Leveraging the exogenous within-neighbourhood variation in salience generated by Islamic terrorism, I show that a single attack can result in an increase in violent crime by around 3.2% (relative to the monthly median) in neighbourhoods with large and visible Muslim communities. Similarly, when salience is measured using the number of newspaper articles ($\approx$ 1.5%) and changes in Google trends searches ($\approx$ 1.1%), estimates remain statistically significant. Notably, the estimated effect rises to 5% for violent crime when attacks happen inside the UK. This baseline result is corroborated using a matched difference-in-differences approach and a battery of robustness checks.

Next, I test for the presence of spillover effects and find that crime rises not only in areas where Muslims live or go to the Mosque, but also in nearby neighbourhoods within a 2.5km radius. This is particularly the case for anti-social behaviour (ASB) which tends to rise most in such adjacent neighbourhoods ($\approx$ 2%). These spillover estimates suggest that the magnitude and spread of anti-Muslim backlash will depend not only on the characteristics of "treated" neighbourhoods, but also on those of the entire local area.

Therefore, I conduct a detailed heterogeneity analysis comparing the magnitude of effects across a number of socio-economic characteristics and the size of losses from austerity. These speak directly to whether hostility towards minorities and immigrants is rooted in deprivation/financial fragility (Fetzer, 2019; Liberini et al., 2019), or social factors such as a lack of contact (Alba et al., 2005; Alrababa'H et al., 2021). As a result, I find that areas with lower levels of education ($\approx$ 11%), low degrees of religious fractionalisation ($\approx$ 19%), lower migrant inflows from 2001-2011 ($\approx$ 16%), but high share of foreign-born residents ($\approx$ 20%) see the largest increases in violent crime. These results suggest that hostility towards Muslims, and by extension other minority groups, can be partly attributed to a lack of contact and exposure, exacerbated by traditional and social media platforms that reinforce pre-existing prejudices (Bursztyn et al., 2019; DeMarzo et al., 2003; Schneider-Strawczynski and Valette, 2025).

Prior work has explored how political preferences, seen in electoral results, can reveal private sentiments and lead to a deterioration of social norms Bursztyn et al. (2020); Clifton-Sprigg et al. (2020). Based on this idea, I use data from the 2010 and 2015 General Elections, and the 2016 Brexit Vote to check whether backlash in areas that voted for right-wing anti-immigrant parties (UKIP/BNP) or for *Leave*, saw larger crime responses to changes in salience. Indeed, I find a large gaps in effects between areas with high vs. low support. However, I go further and consider whether the heterogeneity in these effects is explained by differences in baseline socio-economic characteristics. To do so, following Illing et al. (2024), I use a re-weighting strategy that makes sure that, on average, the characteristics of low-voting areas match those of high-voting ones. This approach reveals that socio-demographic characteristics such as those given above are sufficient to explain most of the gap in effects. Once again, this suggests that anti-minority sentiment is more closely connected to lack of contact and misinformation, than to poorer economic conditions.

This paper relates to several strands of the existing literature. It builds on recent empirical work evaluating the consequences of terrorism on the integration of Muslim communities, voting outcomes, political attitudes or hate crimes (Böhmelt et al., 2020; Deloughery et al., 2012; Gould and Klor, 2016; Hanes and Machin, 2014; Ivandic et al., 2024; Swahn et al., 2003). These find that terrorist attacks deter integration, and strengthen existing hostility from locals, leading to more widespread support for far-right and nationalist parties, and even retaliatory hate crimes. Closest to this paper is Ivandic et al. (2024) that estimates the short-term impact of ten different Islamic terrorist attacks on hate crimes committed in the Greater Manchester area by comparing crime trends across several minority groups. This paper, on the other hand, uses police recorded data from neighbourhoods throughout England and Wales. This not only allows for greater flexibility when it comes to identification, but also provides richer cross-sectional variation for heterogeneity analysis.

More generally, there is also a newly emerging literature documenting the importance of political shocks, such as the election of Donald Trump or the decision of the UK to leave the EU, and the consumption of social media content as determinants of hate crime (Clifton-Sprigg et al., 2020; Müller and Schwarz, 2021, 2023; Rushin and Edwards, 2018). These studies highlight how such electoral results, that reveal the extent of anti-minority sentiment, or "echo chambers" that reinforce existing beliefs can rapidly erode previously held social norms. However, unlike this paper, these studies do not consider the socio-economic conditions that make the emergence of hate crime more or less likely.

Moreover, I contribute to the literature debating the underlying causes of growing anti-minority and anti-immigrant sentiment Alesina and Tabellini (2024). Situated within this are attempts to analyse the determinants of the *Leave* vote in the Brexit referendum, which according to early survey evidence was driven primarily by concerns about immigration (Alabrese et al., 2019; Becker et al., 2017; Goodwin and Milazzo, 2017). It remains unclear whether these concerns were rooted in the worsening economic circumstances or simply misperceptions about migrant communities and their impact on wider society. This paper provides evidence in favour of the latter view that social factors that make individuals prone to misinformation play a more important explanatory role.

The rest of the paper is organized as follows. Section II describes the data for crime, the distribution of Muslim communities around the UK, Muslim salience and the control variables. In Section III, I discuss my identification strategy and robust checks. Section IV presents the results, and finally section V concludes the paper.

2 Data

In this section, I discuss the main data sources used in this article. Before doing so, I briefly discuss some key units in UK statistical geography. In this paper, I use, and often refer to, three non-nested geographic units: Lower Layer Super Output Areas (LSOAs), Westminster Parliamentary Constituencies and Local Authority Districts (LADs).[3] LSOAs are the smallest, with around 1500 residents, and will be referred to as neighbourhoods. Data on LSOAs is at the lowest level disaggregation. Westminster Parliamentary Constituencies (often referred to as simply *Constituencies*) are large units with just over 100,000 residents that are used to elect local Members of Parliaments (MPs) during General Elections. As one focus of our analysis concerns political preferences, we will primarily work with constituencies when performing heterogeneity analysis. Lastly, LADs are the main administrative units in the UK. There are approximately 350 such units that are responsible for local government policies and public good provision. Since LADs are the most recognisable by most residents, these will be used in my treatment definitions.

2.1 Crime

The data on crime at the neighbourhood-level, for 35,750 LSOAs, is obtained from police crime records for England and Wales. These are compiled by 43 Police Force Areas (PFAs) that are responsible for the day-to-day running of policing activities.[4] My raw sample contains information on 15 different types of crime at a monthly frequency for the 2011-19 period.

[3] Whenever another is used it will be noted and explained in a separate footnote.

[4] The raw historical data is available at `https://data.police.uk`. Data from the British Transport Police has been excluded due to potential measurement error associ-

To reduce measurement error from misclassification, these are aggregated into three broader categories: violent (violent & sexual offences, robbery, arson, weapons, public order and disorder incidents), non-violent (anti-social behaviour, substance, burglary, shoplifting, vehicle crime, bike theft, persons theft and other theft) and total crime (the sum of the preceding two categories).[5]

To account for differences in population sizes across neighbourhoods, each type of crime is expressed as the number of incidents per 1000 residents, where population figures are taken from the 2011 Census. Table 1 shows that, on average, just under a third of all reported incidents are violent crimes, with about two being reported each month. Moreover, the distribution of these crime rates is right-skewed, meaning that there is a small number of neighbourhoods in which crime is ubiquitous. For identification, this means that accounting for unobserved differences across neighbourhoods will be important in avoiding endogeneity problems.

Table 1: Summary Statistics - Crime (2011-2019)

Variable	N	Mean	Median	SD
Total Crime	3,752,951	8.080	5.734	11.294
Violent Crime	3,752,951	2.505	1.680	3.416
Non-violent Crime	3,752,951	5.575	3.825	8.546

Notes: This table reports summary statistics for three broad categories of crime: total, violent and non-violent crime recorded at the neighbourhood-level (LSOA) at a monthly frequency for the period 2011-2019.

2.2 Muslim Communities

Information about the religious composition of residents is taken from the 2011 Census which provides data on the share of residents of an LSOA that identify as Muslim. Although Islam is still a minority religion in the UK, the number of adherents has risen considerably over the last two decades

ated with the location of the reported crimes. Moreover, data is not available for all LSOAs that fall into the Manchester PFA after July 2019 due to misreporting.

[5] Due to changes in Home Office Counting Rules, in place from April 2019, the broader categories are not comparable over time. Hence, we discard all observations for from that date. Similarly, due to a large jump in the number of recorded violent crimes, observations for the first eight months of 2011 are also excluded.

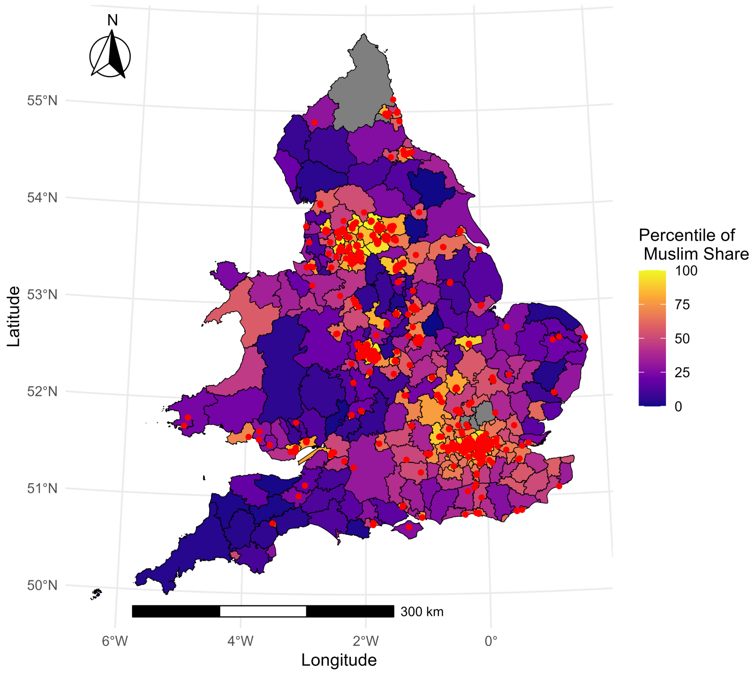
with the share of Muslims in the general population rising from 3% (2001) to 4.8% (2011) to 6% (2021) and the number of mosques (or masjids) growing commensurately. Most of this growth has clustered around older existing communities that provide support networks and aid with employment. Interestingly, the most popular areas for this have been former industrial cities in the North of England, such as Birmingham, Leeds and Sheffield, where many Muslims settled after the Second World War while looking for work. Sadly, many of these areas have recently seen poorer economic performance, often being seen as those left behind, as structural changes reduced the share of manufacturing in the UK economy. [6]

The data on Mosque locations is provided by `https://www.muslimsinbritain.org/index.php` which hosts a definitive directory of all Mosques around the UK. In particular, I use the September 2024 edition of the data, which contains records on 2156 Mosques countrywide. [7] This is because past versions contained significant omissions and would introduce large measurement error into the analysis. Most mosques are located in older community centres or buildings used by the Church of England (Church or parish buildings) that were sold to local Muslim communities. This means that the creation of a new Mosque is almost entirely demand-driven. Therefore, it is natural to use the presence of a Mosque as a visible marker for the presence of a relatively large Muslim community in the area. Figure 1 plots the distribution of Mosques countrywide, alongside a heatmap for the share of self-identified Muslims at the district-level. Unsurprisingly, the two are strongly correlated, and, as noted above, are largest in urban areas, particularly in the North of England.

[6] Importantly, this means that the share of Muslim residents in our analysis, used in part as a measure exposure to salience shocks, will be correlated with other socio-economic characteristics. This is one reason that neighbourhood-specific trends are used in the analysis - to capture this potential source of endogeneity - and are the motivation for one the robustness exercises, discussed below.

[7] Note that this is a subset of the full sample that excludes prayer rooms found in hospitals, student union buildings or airports, as well as multi-faith rooms/centres.

Figure 1: Plot of Mosque Locations and Muslim Share



Notes: A plot of the locations of mosques across England and Wales, alongside a heatmap for the percentiles of the share of Muslim residents at the local authority district-level. The areas in grey lack data due to changes in local authority boundaries.

For the coming analysis, I use the distribution of Mosques and Muslim residents to create a single measure of exposure to changes in salience. In particular, I suppose that a "treated" or "exposed" neighbourhood is one that is in the 75th percentile for the share of Muslim residents in its LAD, and is within 500m of a Mosque. Table 2 is a contingency table summarising treatment variable. About 6% of ll neighbourhoods are sufficiently close to a mosque, but two-thirds of areas with mosques have a relatively large share of Muslim residents. To make sure that the results are not driven by this specific choice, I show that they are robust to several other definitions of exposure.

Table 2: Contingency Table of Mosques and Muslim Share (%)

	Muslim Q75 = 0	Muslim Q75 = 1
Mosque = 0	72.60	21.40
Mosque = 1	2.00	4.00

Notes: This contingency contains the percentage of all neighbourhoods split by whether they are within 500m of a mosque, or whether the share of residents that identify as Muslim in that neighbourhood is in the top quartile of the local authority district.

2.3 Salience

My baseline results will be presented for three related measures of salience. The first, and most straightforward, is simple a binary index for whether a major terrorist attack has occurred in a given month based on the data available in the Global Terrorism Database. To avoid using attacks that would not be noticed by UK residents, I make several sample restrictions.[8] Specifically, I require that any terrorist attacks, attributed to Islamic groups, need to have taken place in the UK, Western Europe or the US, and, for the latter two, to have at least 5 fatalities. As a result, my sample contains 33 Islamic terrorist attacks with most occurring in the 2015-2017 period.

The second measure is based on the number of news articles, from UK-based sources, written about Islamic terrorism in each month as recorded by LexisNexis. This includes articles both online and in-print, as well as local newspapers. To make sure that the articles are relevant, I use the following rule as a filter: "(terror or terrorist or terrorism) & (attack or bomb or bombing or incident) & (muslim or islam or islamist or jihadi or isil or al-qaeda or isis or islamic state)" (Ivandic et al., 2024). As expected, the number of articles is highly responsive to the occurrence of a terrorist attack: on average, in months where such an attack has occurred, the number of articles written jumps from 933 to 2472. Since publishing an additional article is unlikely to generate a substantial response in crime, in the analysis, I express as the number of articles in the thousands.

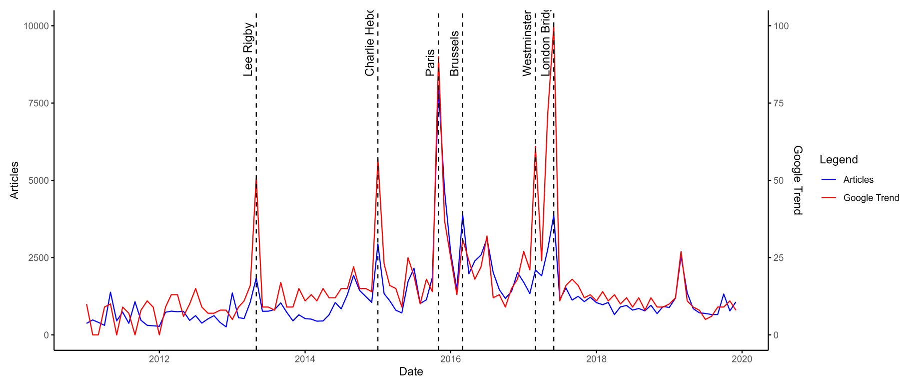
The third measure is based on Google Trends data at the monthly-level. This is a metric of search interest that Google normalises into a 0-100 index relative to the month with the greatest search activity. Specifically, I use the values for the search terms "islamic terrorist". Google search activity

[8] Although these are similar to those used in Ivandic et al. (2024), the

reflects the same pattern as newspaper articles: the average value of the index is about three time higher in months when an attack occurs.

Figure 2 illustrates the time series for number of articles and the Google search index. It is clear that the two are highly correlated, while their largest spikes correspond to major Terrorist attacks. Conceptually, the terrorist attacks are the real-world shocks that underlie any changes to either the number of articles published or Google searches made. The pair, in turn, cause changes to salience. Hence, in the months that terrorist attacks do take place, the salience of Muslim communities will definitely be at its peak. It is then these extreme changes in salience that potentially cause violent backlash against local Muslim communities.

Figure 2: Plot of Muslim Salience



Notes: A plot of different measures of Muslim salience. The first is the number of UK-based newspaper articles written about the topic of Islamic terrorism (identified using several keyword combinations), the second is a relative index for the number of Google searches occurring in the UK about Islamic terrorism. The vertical dotted lines correspond to a selection of major Islamic terrorist attacks.

2.4 Controls

For the heterogeneity analysis and re-weighting exercises, I use a set of control variables at the neighbourhood and constituency-levels from the 2011 Census, available at the NOMIS repository. These characteristics are selected to reflect several dimensions of local development, such as education and unemployment, and the socio-demographic profile of each neighbourhood through measures such as religious fractionalization and share of foreign-born residents. I also include a per person estimate of losses from local austerity cuts, taken from Beatty and Fothergill (2013), as areas subject to deeper austerity cuts may subsequently experienced a greater increase in crime in response to shocks to salience (Giulietti and McConnell, 2020).

In addition, to study the role of political preferences in shaping degree of backlash, I collect data on election results at the constituency-level from the UK Electoral Commission. In particular, I use the results of 2010 General Election, 2015 General Election and the 2016 Brexit Referendum focusing on the share of votes in each constituency made in favour of right-wing or anti-immigrant parties. For the 2010 election, I focus on the combined vote for UKIP (UK Independence Party) and the BNP (British National Party) that, at the time, were the most popular parties advocating for strong restrictions to immigration. However, for the 2015 election, due to the fall of BNP into relative obscurity, I just focus on UKIP that absorbed most of the former's supporters by that time. While for the 2016 Brexit referendum, O use the share of votes going to *Leave*. [9]

Table 3 presents the mean and standard deviation of a a selection of these control variables split by "treated" and "untreated" neighbourhoods. In general, "treated" units were located in constituencies with a slightly lower proportion of votes going to anti-migrant parties. Moreover, these areas were more densely populated, had higher levels of education, were more religious difference, had a higher share of migrants, but were more strongly affected by austerity policies. This suggests that the distribution of visible Muslim communities across England and Wales is non-random. This will be an important source of endogeneity that my identification strategy will account for.

[9] In general, all three variables can be seen as proxies for the extent to which locals hold anti-migrants beliefs. As a results, they are strongly correlated ($\rho \geq 0.7$). Notably this implies, *contra* Fetzer (2019) that such views have roots before the start of any austerity policies.

Table 3: Summary Statistics - Control Variables

Variable	Treated	Untreated
2010 GE	5.198	5.560
	(3.315)	(2.917)
2015 GE	12.267	14.265
	(5.943)	(5.622)
Brexit	49.678	53.324
	(12.247)	(10.946)
Pop. Dens.	89.976	35.429
	(49.973)	(50.998)
Educ.	0.126	0.155
	(0.025)	(0.030)
Relig. Frac.	0.576	0.465
	(0.126)	(0.101)
Foreign	0.349	0.119
	(0.141)	(0.127)
Austerity (GBPpc)	539.876	471.166
	(90.974)	(120.688)

Notes: This table of summary statistics presents the mean and standard deviation (in brackets) for a selection of control variables split by treatment and control units. The variables are: share voting for UKIP/BNP in 2010 general election, share voting for UKIP in 2015 general election, share voting for Brexit, population density, share of residents with at most GCSE-level qualifications, religious fractionalisation, share of foreign-born residents, and austerity losses in GPB per person. The election results are at the constituency-level, the austerity losses are at the district-level, while the rest are at LSOA-level.

3 Empirical Methodology

My goal is to estimate the impact of changes in Muslim salience on the local crime rate. To do this, I exploit the exogenous variation generated by extreme changes in salience induced by major Islamic terrorist attacks. Indeed, if a rise in Muslim salience causes some degree of violent backlash, we should observe a larger increase in violent crime in areas with a visible and relative large Muslim community. To test this idea, I use the following panel regression given in equation 1:

$$y_{it} = \alpha_i + m_t + \tau(\text{Salience}_t \times \text{Muslim}_i) + \beta\text{Salience}_t + \text{trend}_{it} + \varepsilon_{it} \quad (1)$$

where y_{it} is the crime rate in neighbourhood i at time t. To account for baseline time invariant differences across neighbourhoods, as well as monthly fluctuations in crime that are common across all neighbourhoods, I include both neighbourhood (α_i) and month (m_t) fixed effects. Moreover, to capture differences in crime across neighbourhoods, perhaps driven by unobservable time varying differences across areas, I fit neighbourhood-specific quadratic crime trends (trend_{it}). The term Salience_t is a variable measuring the salience of Muslim communities nationally at time t. As discussed above, these are a binary index for when a major terrorist attack has occurred in a given month, the number of articles (in thousands) referencing Islamic terrorism, and a relative measure of the number of Google searches on the same subject. Next, Muslim_i measures whether a given ward is within 500m of a mosque and has a population of Muslim residents in the top 90th percentile of its local authority district. Lastly, ε_{it} is the error term. I cluster standard errors at the neighbourhood-level.

The relevant treatment effect is τ which captures the differences in the way that neighbourhood-level crime responds to changes in salience between areas with a visible Muslim community and those without one.[10] In this ways, the specification in equation 1 bears a resemblance to both a difference-in-difference style design since comparisons are made between treated and untreated areas, and a Bartik-style shift-share design where

[10] Implicitly, I assume that there are no dynamic effects on crime lasting over several months. This is consistent with previous findings such as those in Ivandic et al. (2024) who found that hate crimes directed against Muslims compared to other minority groups spiked for only a few days following an Islamic terrorist attacks. Moreover, although I cannot directly observe the identity of the victims of these recorded crimes, the best explanation, given the context, for any sharp changes in crime rates inside these neighbourhoods is that it is the Muslim communities that are being victimised.

Muslim_i (capturing cross-sectional variation in Muslim communities across England and Wales) measures local exposure to national shocks in salience Salience_t.[11] Nonetheless, for the main specification, the parallel trends assumption is not required as we allow for differential trends in crime via the inclusion of neighbourhood-specific quadratic crime trends. More formally, I require that the composite term ($\text{Salience}_t \times \text{Muslim}_i$) is uncorrelated with any other determinants of the crime rate contained in the error term (ε_{it}). This implies that:

$$E[\text{Muslim}_i \times \varepsilon_{it} | \alpha_i, m_t, \text{Salience}_t, \text{trend}_{it}] = 0 \tag{2}$$

Hence, the biggest threat to identification in this case comes from variables codetermined with the distribution of Muslim communities that are correlated with changes in the crime rate (Adão et al., 2019; Goldsmith-Pinkham et al., 2020). Put another way, these are variables that would cause crime to rise when Muslim salience increases even in the absence of any Muslim residents. If the neighbourhood-specific quadratic time trends are adequately specified, then this assumption is not implausible, yet remains untestable. To bolster its credibility, I conduct a number of robustness checks and falsification exercises aimed at showing that shocks to salience affect only Muslim communities. Moreover, as described below, I use an alternative strategy to show how similar treatment effect estimates can be obtained based on a different set of identifying assumptions.

Corroborative Approach

To show the robustness of our baseline results, I employ a matched difference-in-difference design which directly compares the evolution of crime in treated and matched untreated areas.[12] The main difference to the specification in equation 1 is the presence of a full set of unit (α_i) and time fixed effects (η_t):

$$y_{it} = \alpha_i + \eta_t + \tau(\text{Salience}_t \times \text{Muslim}_i) + \varepsilon_{it} \tag{3}$$

The other variables have the same interpretation as before. Note that now we cannot allow for unit-specific trends, while the time fixed effects absorb

[11] Curiously, although both Müller and Schwarz (2021) and Braakmann et al. (2024) use a similar strategies to my own, the former describes their setup as Bartik-style, while the latter describes it as difference-in-difference-style. This is a puzzling because the model in Müller and Schwarz (2021) has a two-way fixed effects structure used in difference-in-difference designs, while the model in Braakmann et al. (2024) does not.

[12] The model in equation 1 is preferred for all additional analysis because it preserves a larger sample of observations and leads to more precise estimates.

our measures of salience as they vary only in their time dimension.[13] In this case, the average treatment effect on the treated is identified as long as counterfactual trends in crime are parallel between areas with and without visible Muslim communities:

$$E[\text{Muslim}_i \times \varepsilon_{it} | \alpha_i, \text{Terror}_t, \eta_t] = 0 \tag{4}$$

On our full sample of neighbourhoods, this assumption is not plausible as treated units differ substantially in terms of observable outcomes from untreated ones which will potentially imply a different set of crime trends. To tackle this problem, I use a combination of exact and propensity score matching. In particular, I match exactly on the deciles of the average pre-treatment levels of crime, and quintiles of population size and level of education, and subsequently use nearest neighbour propensity score matching on a set of observables at the neighbourhood-level. The covariates used are: population density, proportion of youths (aged 0-24), manufacturing share of employment, share of manual and semi-skilled occupations in total employment, share of residents living in socially-rented accommodation (social housing), gross value added per capita, share in full-time work, share of residents that are male and proportion of residents that are long-term unemployed. Standard errors are clustered at the level of the matched neighbourhoods to account for the variance from the matching step in the regression (Abadie and Spiess, 2022).

Although the conditional parallel trends assumptions, being stated in terms unobservable counterfactual trends, is not directly testable, one can provide evidence in favour of its plausibility. In particular, as the first terrorist attack in the sample occurs in May 2013, it is possible to leverage the 2011-2013 pre-period to verify whether crime trends are indeed parallel between wards (conditional on covariates). Formally, I follow Borusyak et al. (2024) in validating this design by performing a pre-trend test on the sample of untreated observations. To do so, I estimate the following regression by OLS:

$$y_{it} = \theta_t + \eta_i + \sum_{p=-P}^{-1} \gamma_p 1\{t = \tau_{ij} + p\} + \epsilon_{it} \tag{5}$$

[13]Note that in this setup, we do not have differential timing meaning that there is no danger of making forbidden comparisons of the type discussed in the recent difference-in-difference literature (De Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021). For two recent reviews see Roth et al. (2023) and De Chaisemartin and d'Haultfoeuille (2023).

where y_{it} is the outcome variable, θ_t is the time fixed effect, η_i is the unit fixed effect, p denotes some number of months before the first terrorist attack in relative time, τ_{ij} is the date of the first attack and ϵ_{it} is the error term. All periods more than 6-months before the first attack are used as the base period.

The results of this test are provided in figures 8 - 10 in the Appendix, where a visual inspection reveals that virtually all point estimates are close to zero and do not exhibit any trends or patterns. For the 1:1 matched sample, the entire set of estimates is not statistically significant at the 5% level. For the 1:3 and 1:5 matched samples all but the final point for total and non-violent crime, corresponding to April 2013, are equally not significant. The reason for this jump, driven by non-violent crime, is unlikely to be an anticipation effect, but may reflect increasing tension after the the murder of Mohammed Saleem by a white supremacist, who additionally planted three bombs in nearby Mosques, in an attempt to provoke racial conflict.[14] Overall, this provides some evidence that the parallel trends assumption is plausible after the matching.

Additional Analysis

In addition to our baseline results, we consider whether neighbourhoods close to treated areas also see an increase in crime. To check this we estimate equation 6 where neighbourhoods are divided into 500m based on their distance from a treated area. Since I use 10 such bins, the maximum distance considered here is 5000m:

$$\begin{aligned} y_{it} = \alpha_i + m_t + \tau(\text{Salience}_t \times \text{Muslim}_i) \\ + \sum_{j=1}^{10} \gamma_j(\text{Dist}_{ij} \times \text{Salience}_t) + \beta\text{Salience}_t + \text{trend}_{it} + \varepsilon_{it} \quad (6) \end{aligned}$$

Most variables have the same interpretation as before. However, the variable Dist_j is a binary indicator for whether a given neighbourhood i belongs to bin j, and the parameter γ_j measures the effect of changes to Muslim salience for each distance.

[14] More information on this can be found at the Guardian newspaper: `https://www.theguardian.com/society/2021/apr/05/daughter-of-murdered-muslim-man-calls-for-official-islamophobia-definition?CMP=share_btn_url`

Furthermore, I document the heterogeneity of treatment effects across the characteristics of the terrorist acts themselves as well as the socio-economic characteristics of surroundings areas. To start with, I compare the magnitude of retaliatory responses based on whether the terrorist act occurred in the UK or overseas. The intuition being that locals may feel more threatened if attacks are targeted at and hurt members of their community. Next, I explicitly evaluate the role of austerity, which has often affected disadvantaged communities the strongest potentially leading to higher out-group violence, by comparing crime trends between areas in the upper and lower quartiles in terms of per person losses from welfare cuts. I follow a similar approach for other socio-economic indicators such as education, unemployment or share of foreign-born residents (at the parliamentary constituency level).

Lastly, I consider the role of political affiliation by checking whether neighbourhoods located in constituencies with a higher share of votes going to right-wing parties (in the 2010 and 2015 general elections, as well as during Brexit) saw a greater rise in violent crime. Moreover, I extend this analysis by using a re-weighting strategy (explained in more detail below) to determine whether the gap in estimated effects can be explained using social or economic observables.

Robustness Checks

As noted above, I conduct a battery of robustness checks to bolster the plausibility of my baseline identification strategy. To start with, I consider whether the results are simply driven by the definition of a treated area. Hence, I modify the treatment definition to: (i) whether the share of Muslim residents is in the 90th percentile for the local authority district, and (ii) whether the share of Muslim residents is an outlier, defined as a share greater than $1.75 * \text{IQR}$ (where IQR is the interquartile range), for the local authority district.

As an important falsification exercise, I consider whether areas with large populations of other religious minorities, namely Jews, Hindus and Sikhs, experience a similar growth in crime during times of higher Muslim salience. We expect that these effects should be either not statistically significant or, at most, slightly negative, due to spatial displacement effects. To further alleviate worries that the effects are driven by trends in factors correlated with the presence of Muslim communities, I check whether the baseline estimates are robust to the inclusion of a set of interactions between measures of salience and various neighbourhood-level observables. Although some of these may be mediators, it is expected that estimated coefficients should have a similar magnitude after the addition of these controls.

Finally, I conduct a placebo check to evaluate whether the estimates are driven by a higher random shocks or the higher responsiveness of crime in treated area. Intuitively, if this is the case then, on average, any random series of shocks would yield large and positive coefficient estimates. Therefore, I simulate 1000 series each with 33 placebo terrorist attacks, and re-estimate equation 1 for each one collecting together the resultant set of coefficient estimates. If treated areas are uniquely more responsive then the average of these estimates should be statistically significant and positive.

4 Results

4.1 Muslim Salience and Crime

Baseline Estimates

The baseline estimation results for the effect of changes to Muslim salience, induced by terrorist attacks, on total, violent and non-violent crime rates are presented in Table 4 below.[15] As noted above, for each crime rate, three different measures of Muslim salience are used: a binary indicator for major terrorist attacks, the number of newspaper articles written about Islamic terrorism and a relative measure of the number of Google Searches that link Islam and terrorism. The first reflects the actual events that make Muslim communities more salient to the general public, while the last two are more noisy reflecting the supply and demand of information respectively. Apart from illustrating two potential channels through which salience leads to action, the inclusion of these additional measures acts as a robustness check for whether any changes in crime are really driven by acts of terrorism or other national-level shocks.

Across all specifications there is a significant increase in the total crime rate in treated areas. Relative to its monthly median, this corresponds to an increase of between 0.5% - 2% with the actual terrorist attacks series measuring the greatest effects. This jump is primarily driven by violent crime that rises by up to 3.2%. Non-violent crime, however, appears to be

[15] In interpreting these coefficient estimates, I will commonly refer to percentage changes in the each crime rate. Unless stated otherwise, these will be relative to the *median* of each monthly crime rate at the neighbourhood-level.

largely unaffected and is only weakly significant in one specification. The coefficients for articles and google trends are also positive, yet the former often exceeds the latter suggesting that news coverage, and hence the supply of negative information, is more important for violent behaviour than demand. Interestingly, for both violent and non-violent crime, each salience series itself is also significant indicating that crime rises across untreated areas as well - the increase in exposed areas is nearly twice as large! This may be driven by neighbourhood spillover effects, a possibility that I take up in the next section.

Table 4: Impact of Muslim Salience on Crime

	Tot	Tot	Tot	Viol	Viol	Viol	NViol	NViol	NViol
Attack x Treat	0.106***			0.055***			0.051*		
	(0.036)			(0.019)			(0.028)		
Attack	0.115***			0.075***			0.040***		
	(0.006)			(0.003)			(0.004)		
Articles x Treat		0.031**			0.025***			0.005	
		(0.014)			(0.007)			(0.011)	
Articles		0.027***			0.029***			-0.001	
		(0.002)			(0.001)			(0.002)	
Google x Treat			0.020**			0.019***			0.001
			(0.008)			(0.004)			(0.006)
Google			0.026***			0.020***			0.006***
			(0.001)			(0.001)			(0.001)
N	2745211	2745211	2745211	2745211	2745211	2745211	2745211	2745211	2745211

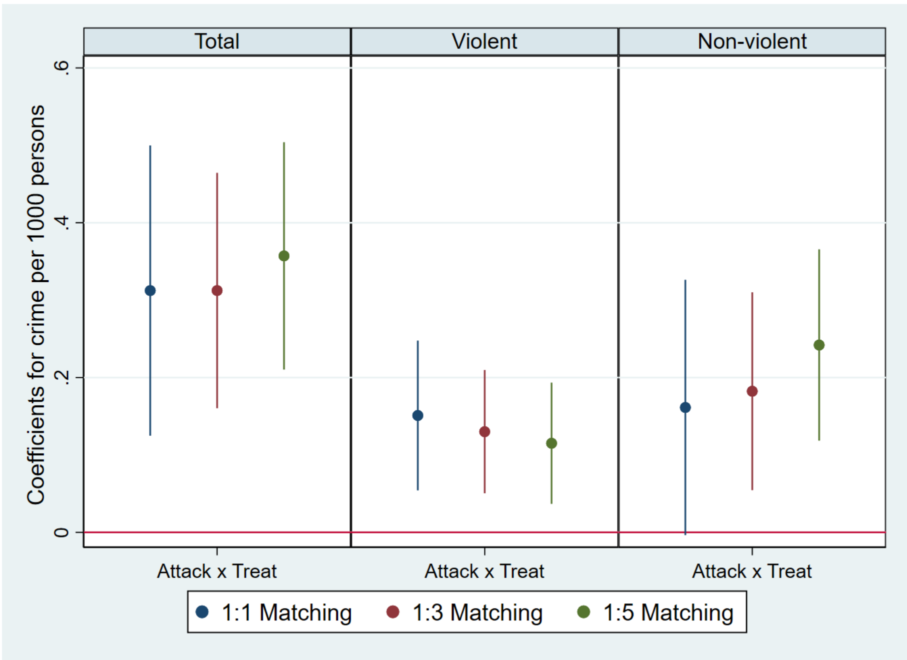
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to Muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. For each type of crime, three measures of salience are used: terrorist attacks, newspaper articles and the Google Trends search index. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Difference-in-Difference Estimates

To corroborate these baseline estimates, I use a matched difference-in-difference approach that compares the evolution of crime between "exposed" and "matched" neighbourhoods. The results of this for 1:1, 1:3 and 1:5 matched samples is presented in Fig 3 using the terrorist attack measure of salience. I focus on this measure because it is my preferred metric, used for the rest of the analysis, and because it clearly delineates a pre- and post-treatment period.

Figure 3: Matched DID Estimates Plot



Notes: A plot of estimates for the effect of Muslim salience on crime crime split based on a matched difference-in-difference approach. Results are shown for three matched samples: 1:1, 1:3 and 1:5, and use the binary index of terrorist attacks to measure salience. Estimates include a 95% confidence interval.

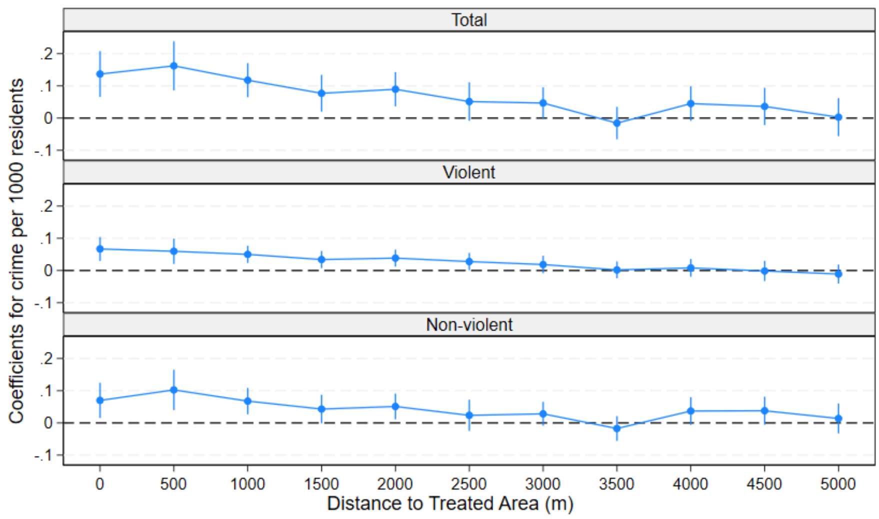
As in Table 4, all effects are positive and statistically significant, though the estimates in Figure 3 are larger, but also less precise. This is because the matched samples are typically smaller compared to those used in the baseline estimation. It is noteworthy that as the number of control units expands, potentially introducing units that are less comparable to treated ones, the coefficients for violent crime fall, while those on non-violent crime rise. The implication of this is that violations of parallel trends, in this context, are likely to be associated with under-estimates of the true effect.

The full set of results and the estimates for the other measures of salience can be found in Tables 6 to 8 in the Appendix, and are also consistent with the baseline estimates described above.

4.2 Neighbourhood Effects

To measure neighbourhood spillover effects, we consider how the impact of salience differs by the distance of neighbourhoods to the treated areas. These estimates are plotted in Figure 4, and, in general, show that treatment effects diminish with distance, whereby most coefficients after the 2500m are no longer significant. The one exception to this is non-violent crime which rises the most about 500m away from the treated areas. One possible explanation of this difference is that whereas violent crimes are typically intentional in that a perpetrator has a clear target in mind, non-violent crimes are less specific. For example, an assault against members of the Muslim community would usually involve actively seeking a victim, while vandalism or verbal abuse, examples of anti-social behaviour, are much more likely to occur on the high street and not directly where Muslims would live or gather for prayer.

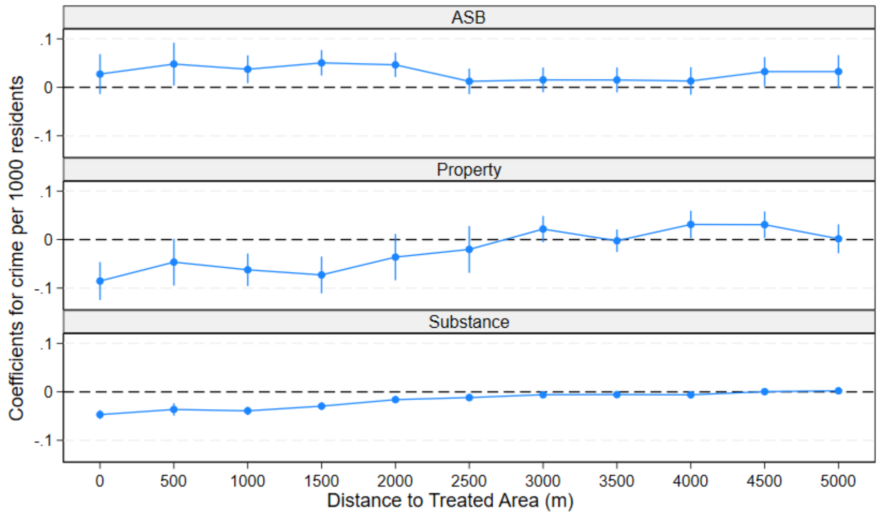
Figure 4: Neighbourhood Spillovers - Muslim Salience and Crime



Notes: An plot of estimates for the effect of Muslim salience on total, violent and non-violent crime split by distance from treated units.

To test this possibility, I separate non-violent crime into its three main components: anti-social behaviour (ASB), property (consisting of burglary, shoplifting, vehicle crime, bike theft, persons theft and other theft) and substance crime. Intuitively, if the above argument is correct, one would expect there to be a positive impact on anti-social behaviour in areas around but not directly in the treated neighbourhoods. The results of this extension are presented in Figure 5. There I find evidence of changes to crime composition: both property and substance crime appear to decline in treated areas. *A priori* the mechanism behind this is unclear, but may be related to changes in police deployment that occur in response to a rise in violence. Nonetheless, as expected, a small but positive effect on ASB can be observed up to the 2500m mark at which it disappears.

Figure 5: Neighbourhood Spillovers - Muslim Salience and Crime



Notes: An plot of estimates for the effect of Muslim salience on anti-social behaviour, property and substance crime split by distance from treated units.

These findings suggest that to capture the full effect of changes to salience on crime, one must include not only the treated areas themselves where Muslim communities may live, but also areas nearby. For this reason, we modify our baseline specification and include an additional treatment indicator for neighbourhoods located at within 2500m of a treated unit. The results from this adjusted specification are given in Table 5.

Table 5: Impact of Muslim Salience on Crime with Neighbourhood Effects

	(1) Tot	(2) Viol	(3) NViol	(4) ASB	(5) Prop.	(6) Subst.
Attack x Treat	0.130***	0.065***	0.065**	0.022	-0.090***	-0.046***
	(0.036)	(0.019)	(0.028)	(0.021)	(0.020)	(0.005)
Attack x Neigh.	0.087***	0.039***	0.048***	0.033***	-0.054***	-0.025***
	(0.014)	(0.007)	(0.011)	(0.007)	(0.010)	(0.002)
Attack	0.090***	0.064***	0.026***	0.097***	-0.052***	-0.024***
	(0.006)	(0.003)	(0.005)	(0.003)	(0.003)	(0.001)
N	2745211	2745211	2745211	3742913	3742913	3742913

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent, non-violent, ASB, property and substance crime defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

In most cases, this appears to increased the magnitude of the estimated causal effect as the areas impacted by the spillovers were formerly part of the control group, and are more that twice as large as "unexposed" areas. Moreover, as expected, for most types of crime, the neighbourhood spillover effects are smaller than those for directly treated units. The one exception to this is the result for ASB (anti-social behaviour), where only the spillover effect is statistically significant.

4.3 Heterogeneity Analysis

The spillover-adjusted estimates lead to two further questions. If distance to exposed areas matters, is the distance to the terrorist attack itself important? Moreover, since the effects are far-reaching, do they depend on the type of neighbourhoods that live nearby to treated ones? To answer this, I conduct heterogeneity analysis by the location of the terrorist attacks and across various socio-economic characteristics at the constituency-level. [16]

[16] The one exception to this is austerity where, as noted above, I use data from Beatty and Fothergill (2013), which is only available at the local authority district-level.

4.3.1 Attack Location

Table 9 in the Appendix presents the estimates of the impact of Muslim salience on crime split by whether the changes in salience are due to terrorist that happened in the UK or abroad. For violent crime, the pattern is clear: domestic attacks are far more important leading to a just under 5% increase in the violent crime rate relative to the monthly median. The inclusion of controls for spillover effects increases this estimate to 5.5%. Underlying these results is the idea, similar to that of in-group bias, that attacks "closer to home" are perceived to be more threatening than those occurring abroad, hence generating stronger backlash (Shayo, 2020). However, this apparent pattern does not hold for non-violent crimes, where attacks abroad matter a lot more causing an jump of around 2%. This asymmetry can best be explained by differences in frequency and severity between attacks that happened in the UK and abroad. Specifically, UK attacks were less frequent but had a higher average number of casualties. As such, international attacks would not create such strong negative reactions among locals, leading to less severe forms of backlash.

4.3.2 Austerity

An recent and important literature has found a robust link between austerity policies and local crime (Bray et al., 2024; Facchetti, 2025; Giulietti and McConnell, 2020; Villa, 2024). Moreover, it is possible that those most strongly affected by such policies would attribute some portion of their economic woes to migrants and other minorities, leading to stronger backlash amongst those with pre-existing prejudices against Muslims.[17] Hence, in Table 10, I re-estimate the adjusted baseline model split results by whether the per person losses from austerity policy (GBP) at the LAD-level are below the 25th or above 75th percentile.

In this case, the effects on both violent and non-violent crime move in the same direction with areas more strongly affected by austerity cuts seeing quantitively larger growth in crime. In particular, neighbourhoods in the top quartile for austerity-related losses see violent and non-violent crime increases by just over 4% and 2% respectively. On the other hand, for areas in the bottom quartile, the estimated effects are not significant. This provides evidence in support of conflict theory that predicts that austerity policies

[17] A similar idea is proposed in Fetzer (2019) as an explanation for the results of the 2016 Brexit vote that resulted in the UK leaving the EU. Notably, this mechanism is suggested in Guriev and Papaioannou (2022) as an important explanation for the rise of populism across many European economies.

(or simply those that increase deprivation) can work to undermine social cohesion by increasing the exposure of local communities to economic shocks and intensifying competition between different social groups (Anderson et al., 2020; Bray et al., 2024). It is also consistent with previous empirical analysis linking worsening economic conditions and anti-minority crimes (Dustmann et al., 2011; Falk et al., 2011; Krueger and Pischke, 1997).

4.3.3 Socio-economic Characteristics

Next, I consider the the differences in responses based on a range of socio-economic characteristics. For the main part, the discussion will focus on violent crime as the differences for non-violent crime are small for both economic and social factors.[18] Tables 12 and 15 present the estimates for neighbourhoods in the bottom or top quartiles of the distribution of a given socio-economic characteristic.

I start with economic factors presented in Tables 12. Columns (1) and (2) consider the role of population density which is a rough proxy for the degree of urbanisation of a constituency. Although both coefficients are weakly significant, the effect in areas of low population density is many magnitudes higher implying an increase of crime by up to 18%! However, one must be careful with this interpretation because the standard error is 5 times greater for this estimate than for its high population density counterpart. Given this imprecision, it is likely that there are very few large Muslim communities outside urban areas. Nonetheless, the larger magnitude could reflect crime patterns in suburban areas with high degrees of out-migration, where the degree of deprivation increases the propensity of locals for committing hate crimes.

Columns (3) and (4) instead look at differences in education: the share of residents with at most GCSE-level qualifications. In areas where educational attainment is lower, the estimated effect is large and significant at just over 11%. This result is in line with previous findings that show that lower education is associated with a less tolerant and more negative attitude towards migrants and other minorities (Margaryan et al., 2021), as well as a greater propensity towards crime (Bell et al., 2022). In the UK context, for example, individuals and areas with lower attainment were more likely to support anti-immigrant parties and vote for Brexit (Alabrese et al., 2019; Becker et al., 2017, 2018). Therefore, it is not surprising that negative attitudes that arise following a terrorist attack can, in such areas, also in-

[18] This will also imply that the results for crime overall are driven by violent crime patterns.

crease the incidence of violent crime. The remaining estimates for long-run unemployment and social housing are at best only weakly significant. It must be noted however that the point estimate is larger for constituencies with higher shares of those in long-term unemployment and lower shares of residents living in social housing.

Moving on to social factors, presented in Table 15, a pattern begins to emerge with respect to the profile of the constituency with the largest effects. As columns (1), (4) and (5) show, the effects on violent crime are largest in places lower degrees of religious fractionalisation ($\approx 19\%$), high share of foreign-born individuals ($\approx 20\%$) and a lower change in the share of foreign-born residents over the preceding decade ($\approx 16\%$). Although estimated again with lower degrees of precision, the significance and consistency of effects is suggestive. In particular, the effects imply that the backlash against Muslims is not necessarily based on new migrant arrivals, but often against already established communities that are perhaps perceived in a different way. Part of this has to do with domestic responses to terrorist attacks that often lead to increasing costs of assimilation (Gould and Klor, 2016), but also how political leaders and news agencies can act to normalize more xenophobic language and willingness to express hostile views against minorities (Bursztyn et al., 2020). This has been exacerbated in the 2010s through the rise of social media which aided in the creation echo-chambers and spread of misinformation particularly among ethnically and religiously homogenous groups sometimes resulting in outward hostility towards migrants and ethnic/religious minorities (Müller and Schwarz, 2021, 2023).

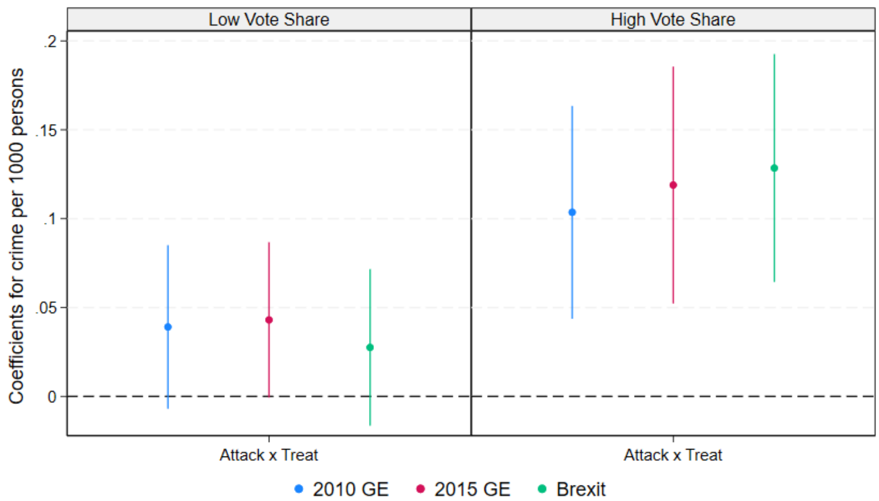
Overall, one can identify several key patterns based on the differences in effect sizes across characteristics. Constituencies that, on average, see lower educational attainment, higher rates of unemployment, fewer migrant inflows, and lower religious diversity see stronger reactions of residents against Muslim communities. To link these findings to previous literature, I now consider the role of political preferences in shaping this backlash.

4.4 Political Preferences

The degree of backlash against Muslim communities strongly depends on the characteristics of the local constituency, often being associated with greater degrees of homogeneity, deprivation and isolation. If so, then this should also be reflected in the revealed preferences of local residents through the voting outcomes of general elections or referenda (Colussi et al., 2021). However, it is not clear whether it is political parties and local leaders themselves that aid the spread of hateful ideas by promote agendas harmful to minorities/migrants, or whether it is the socio-economic circumstances that create tension and lead to greater voting for parties hostile to migrants/minorities? This question touches upon both whether support for populism is ultimately due to social or economic factors, and what policies are available to the government to promote long-term integration and social cohesion.

Hence, to confirm that association between political preferences and violent backlash, I consider the results of three elections the 2010 General Election, 2015 General Election and the Brexit Vote. Specifically, I check whether violent crimes responds more strongly in areas whose share of support for right-wing (UKIP & BNP) or leaving the EU was above the median. The results of this exercise are illustrated in Figure 6 below.[19]

Figure 6: Unadjusted Political Preference Gap for Violent Crime



Notes: A plot of estimates for the effect of Muslim salience on violent crime crime split by whether the share of votes going to right-wing political parties or for Brexit was higher or lower than the median. Estimates include a 95% confidence interval.

In each case, there is a large gap in responses that appears to grow over time suggesting that political beliefs and attitudes play an increasingly important role in determining the strength of violent backlash. Indeed, constituencies where the share of votes for Brexit was above the median, saw violent crime jump by nearly 8% in response to a terrorist attack,

[19]The full set of results including those for the re-weighting exercise, discussed, below, are presented in Tables 17 - 19 in the Appendix.

while for those with shares below the median saw positive but not statistically significant effects. This complements the evidence in Clifton-Sprigg et al. (2020), who find that a large 15-20% increase in hate crimes after the Brexit referendum. More generally, these results highlight how the progressive normalisation of private prejudices can increase willingness to express these views violently against minorities Bursztyn et al. (2020); Rushin and Edwards (2018).

Nonetheless, as noted above, the specific reasons for why this gap arises are not apparent. For example, it could be that economic deprivation or cultural homogeneity that explain both the observed voting patterns and the degree of backlash between neighbourhoods in respond to changes in salience. In others, can this treatment effect heterogeneity be explained by compositional differences in the characteristics of the different constituencies and can the contributions of different types of variables be identified? If so, this information could provide governments with the tools to promote greater social cohesion without sacrificing diversity.

Although neither question can be fully resolved, I attempt to provide some suggestive evidence in support of one particular response. To do so, I draw on the work of DiNardo et al. (1996) and more recently Illing et al. (2024) in calculating a "composition-adjusted gap" in treatment effects between neighbourhoods located in constituencies whose share of votes was above/below the median for right wing parties (2010), UKIP (2015) and Brexit (2016), all through a re-weighting strategy. As suggested above, the difference in estimated effects observed above could be explained by differences in the distribution of observable outcomes between the two types of constituencies. To overcome this challenge, I reweigh the constituencies that voted against right-wing parties/UKIP/Brexit (*low voting*) to match the distribution of observables in the constitutes that voted for them (*high voting*). I do so, however, in two steps first by using only social factors related to migration and fractionalisation, and only subsequently adding economic factors. This way the contribution of either group of observables can be separated.

Formally, this approach can be summarised following Illing et al. (2024). Suppose that τ is the treatment effect for a the impact of a terrorism-induced change to Muslim salience on violent crime for any given neighbourhood. Then the raw unadjusted gap, observed above, can be written as the difference in average treatment given the corresponding constituency's voting behaviour:

$$Gap_{unadj} = E[\tau|\text{High}] - E[\tau|\text{Low}] \tag{7}$$

Now consider a vector of covariates X which are potential determinants of the size of the gap and vary between constituencies. In order to account for differences in composition, we would like to find the treatment effect of neighbourhoods in low voting constituencies adjusted to the covariate distribution of high voting ones:

$$E[E[\tau|\text{Low}, X]|\text{High}] \tag{8}$$

To achieve this, again following Illing et al. (2024), I use a re-weighting function, $\phi_x(x)$, to map the characteristics of low to high voting constituencies. This then yields a composition-adjusted gap:

$$Gap_{adj} = E[\tau|High] - \int_{\mathcal{X}} E[\tau|Low, X]dF_X^{Low}(x)\phi_x(x) \tag{9}$$

In practice, I use a probit model to estimate the propensity score, given a set of social and economic observables, of each constituency having a high or low vote share for right-wing parties, UKIP or Brexit.[20] The social characteristics used for the re-weighting are: proportion of male residents, proportion of residents aged 0-24, religious fractionalisation, proportion of foreign-born residents, change in proportion of foreign-born residents over the preceding decade, share of residents born in Africa or the Middle East. The economic characteristics are: population density, manufacturing share of employment, proportion of residents with at most GCSE-level educational qualifications, proportion of residents working in unskilled or semi-skilled occupations, share of residents living in socially-rented accommodation (social housing), share of residents in full-time work, gross-valued added per capita, and the long-term unemployment rate.[21]

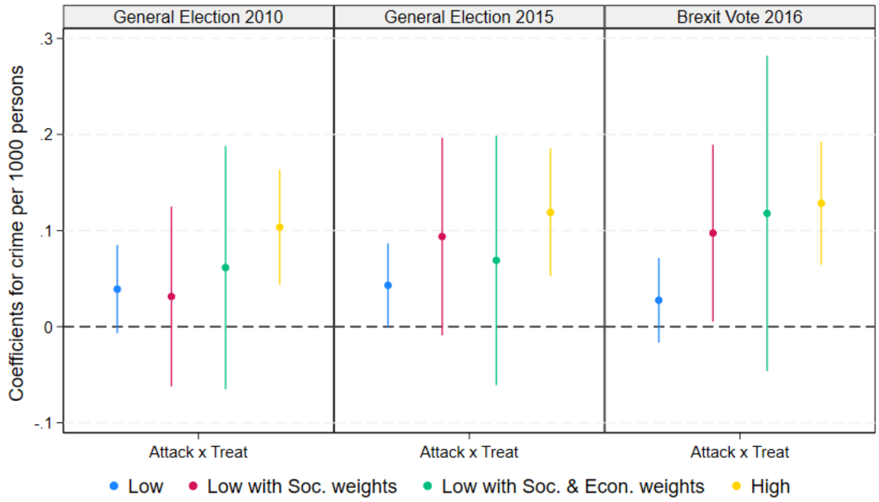
Neighbourhoods in high-voting constituencies are assigned a unit weight, while those in low voting constituencies receive a weight of $\frac{\hat{p}}{1-\hat{p}}$, where $\hat{p}$ is the estimated propensity score. To account for the uncertainty generated by this first step, standard errors are clustered at the constituency-level.

[20] Note that the propensity scores are estimated at the constituency-level. This was primarily due to data limitations as information on voting at lower levels of disaggregation is not available. Unfortunately, this means that standard errors of the resulting estimates will be large.

[21] One exception to this is the the 2016 Brexit Referendum where the proportion of residents with at most GCSE-level educational qualifications is not included as a predictor as it separates out *Leave* and *Remain* constituencies creating potential problems for overlap.

The results of this re-weighting exercise are presented in Figure 7. Although the the estimate are not very precise, an important conclusion can still be gleaned from the results: most of the gap in backlash across political preferences can be explained by social and demographic factors. Notably, the inclusion of economic variables does not change the point estimates very much, and, in the case of the 2015 General Election, even deepens the gap!

Figure 7: Adjusted Political Preference Gap for Violent Crime



Notes: A plot of estimates for the effect of Muslim salience on violent crime crime split by whether the share of votes going to right-wing political parties or for Brexit was higher or lower than the median. Estimates include a 95% confidence interval.

Overall, I find that differences in political preferences matter for the degree of backlash occurring against Muslims during periods of high salience. In particular. neighbourhoods within constituencies with high degrees of support for parties associated with greater hostility towards migrants and minorities see much larger effects on violent crime. However, much of this gap can be explained by differences in social and demographic characteristics such as the share of foreign-born individuals and degree of religious fractionalization. This provides some evidence in support for the "contact hypothesis", whereby greater exchange of ideas and more contact with members of other communities can reduce prejudices and promote integration (Allport, 1979). Indeed, my findings are consistent with recent evidence by Bursztyn et al. (2024), who find that longer exposure to Arab Muslims in the US reduces prejudices, as well as support for political parties and leaders hostile to them. The same mechanism appears to underpin the differences in effects found based on voting preferences in the UK.

4.5 Robustness Checks

In this final section, I discuss the results of each robustness exercise. To start with, Table 20 re-estimates the baseline model with a different set of definitions for a treated area. As described above these are whether the share of Muslims in a neighbourhood is in the 90th percentile for its LAD, and whether the share of Muslims in an neighbourhood is an outlier for that LAD (where outlier is understood as 1.75 times the interquartile range). The first column for each type of crime provides the baseline estimates for reference. The first alternative measure is broader and hence the estimate is expectedly smaller for each type of crime, while the second, being more restrictive, sees larger effects. Nonetheless, for each type of crime, the results are completely consistent with the ones found for my preferred definition.

The next robustness check presented in Table 21 considers whether other religious groups are similarly affected by changes to Muslim salience. Indeed, this partly acts as a falsification exercise as my proposed interpretation of the estimated effects rests on the idea that changes to Muslims salience affect crime against specifically Muslim communities. Finding positive effects for other religious groups would imply that the crime rises not because of the presence of Muslims *per se*, but due to other variables. Fortunately, with the exception of non-violent crime for Hindus, I find no statistically significant effects. The small negative result for Hindus could either have arisen by chance, or perhaps been a result displacement.

To check more thoroughly for whether my results are driven by the actual presence of Muslim communities or other codetermined variables, I add interaction terms between a set of neighbourhood-level observables to the spillover adjusted model. The results of this exercise are given in Table 22. If my baseline results are indeed explained by the endogeneity of the shares, then the inclusion of these covariates should significantly reduce the magnitude of my estimates. However, the opposite actually occurs! Although the point estimate for violent crime is very close to its baseline quantity, the effect of non-violent crime becomes quantitatively larger. Nonetheless, these results show that my original results are not driven by observables that are potentially codetermined with the share of Muslims living in a given neighbourhood. This provides some evidence that the same can be said for unobserved variables as well.

Finally, to make sure that my results are not driven simply by a greater propensity towards violent crime in response to any external shocks, I find the average of a set of estimates for a random simulated series of 1000 "terrorist attacks". If treated areas are systemically more responsive to any shocks, then the average of these estimates should differ from zero. Fortunately, as the results from Table 23 suggest, the average effect is not statistically different zero for each type of crime, and does not overlap with my baseline results either.

5 Conclusion

In this paper, I use neighbourhood-level crime data to investigate how terrorism-induced changes in Muslim salience affect local crime rates. Previous research has used more limited geographic variation, leveraging differences in crime trends across groups of potential victims (Ivandic et al., 2024). Hence, a key contribution of this paper is detailed exploration of treatment effect heterogeneity through the use of country-wide police data. Understanding which factors make violent backlash more or less likely is crucial from a policy perspective, particularly when promoting integration of religious minorities.

Using a Bartik-style model, corroborated using a matched difference-in-difference design, I find that in the same month that a Islamic terrorist attack occurs, violent crime, in areas large and visible Muslim communities, rises by up to 3.2%. Crucially, this depends on the characteristics of nearby communities. For example, in districts hit harder by austerity policies, the estimated effect rises to around 4%. However, the strongest effects are seen in constituencies with lower educational attainment ($\approx 11\%$), lower degrees of religious fractionalisation ($\approx 19\%$) and the lowest size of migrant inflows over the preceding decade ($\approx 16\%$). These results complement recent evidence on the role of social media for driving hate crimes (Müller and Schwarz, 2021, 2023), as well as on the consequences of austerity policies (Bray et al., 2024) and the Brexit referendum (Clifton-Sprigg et al., 2020).

Moreover, these findings have a clear set of public policy implications. Firstly, external threats, perceived or otherwise, can create strong tension between different ethnic and religious communities even in formerly well-integrated countries such as the UK. Secondly, such tension often emerges in the form of violent backlash that has large spillover effects across neighbourhoods. Lastly, to sustain an agenda of social cohesion, local governments should work to promote the exchange of ideas and wider contact across different ethnic/religious groups, preventing the formation of closed and isolated communities in which prejudices can form.

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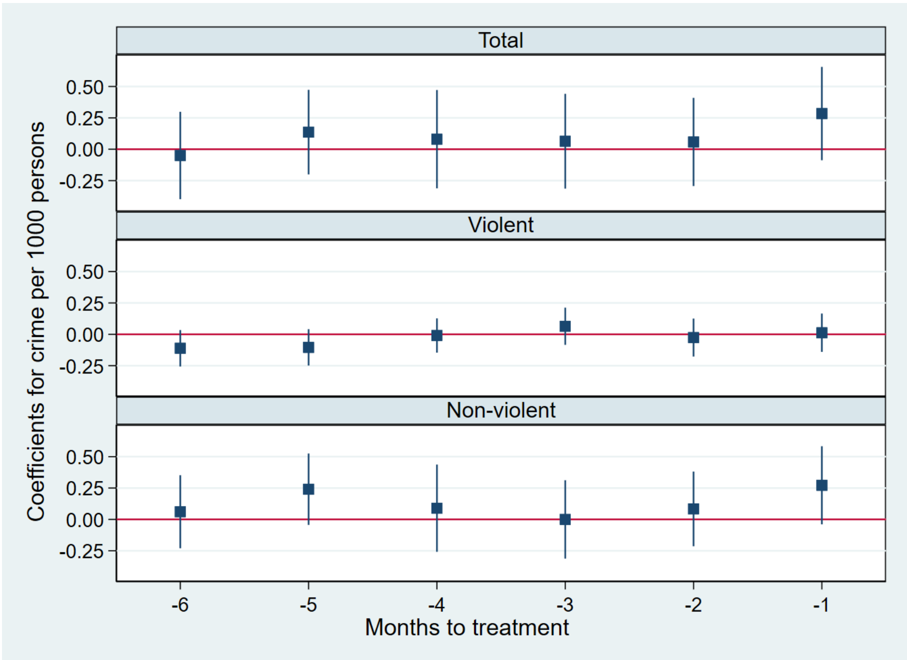
Appendix

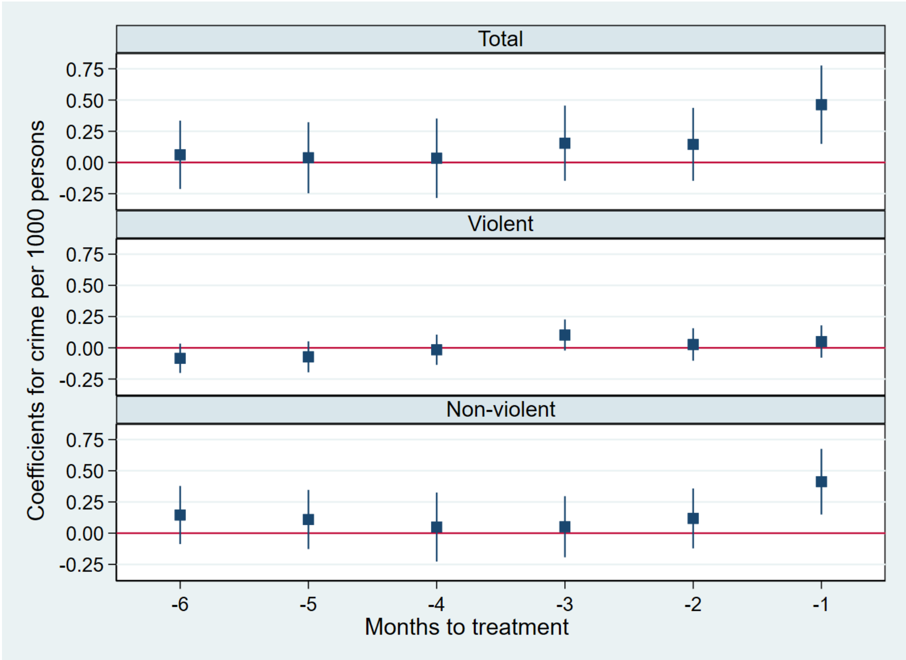
Figures

Figure 8: Pre-trend Estimates for 1:1 Matched DID



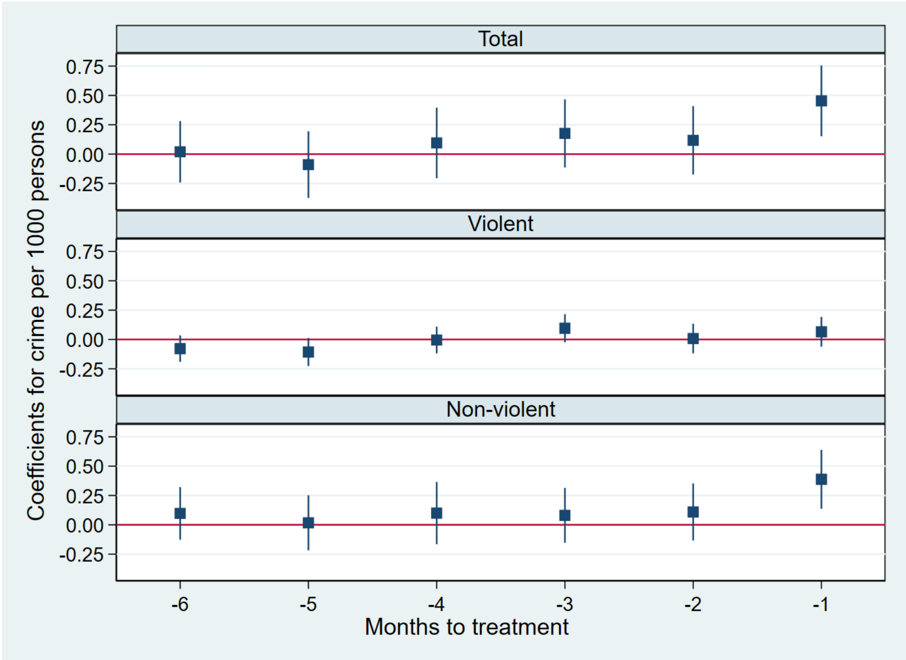
Notes: A plot of pre-trend estimates using the approach suggested in Borusyak et al. (2024) on the 1:1 matched sample. Estimates include a 95% confidence interval.

Figure 9: Pre-trend Estimates for 1:3 Matched DID



Notes: A plot of pre-trend estimates using the approach suggested in Borusyak et al. (2024) on the 1:3 matched sample.

Figure 10: Pre-trend Estimates for 1:5 Matched DID



Notes: A plot of pre-trend estimates using the approach suggested in Borusyak et al. (2024) on the 1:5 matched sample.

Tables

Table 6: Matched DID Estimates of the Impact of Muslim Salience on Crime (1:1)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Tot	Tot	Tot	Viol	Viol	Viol	NViol	NViol	NViol
Attack x Treat	0.312***			0.151***			0.161*		
	(0.096)			(0.049)			(0.084)		
Articles x Treat		0.121***			0.038**			0.083**	
		(0.038)			(0.016)			(0.034)	
Google x Treat			0.076***			0.031***			0.045**
			(0.021)			(0.009)			(0.018)
N	213117	213117	213117	213117	213117	213117	213117	213117	213117

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the matched difference-in-difference estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effects. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. Note that the models use a 1:1 matched sample for each type of crime. Standard errors are clustered at the level of the matched neighbourhoods.

Table 7: Matched DID Estimates of the Impact of Muslim Salience on Crime (1:3)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Tot	Tot	Tot	Viol	Viol	Viol	NViol	NViol	NViol
Attack x Treat	0.312***			0.130***			0.182***		
	(0.078)			(0.041)			(0.065)		
Articles x Treat		0.140***			0.035***			0.105***	
		(0.030)			(0.013)			(0.025)	
Google x Treat			0.084***			0.029***			0.055***
			(0.017)			(0.008)			(0.014)
N	400656	400656	400656	400656	400656	400656	400656	400656	400656

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the matched difference-in-difference estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effects. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. Note that the models use a 1:3 matched sample for each type of crime. Standard errors are clustered at the level of the matched neighbourhoods.

Table 8: Matched DID Estimates of the Impact of Muslim Salience on Crime (1:5)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Tot	Tot	Tot	Viol	Viol	Viol	NViol	NViol	NViol
Attack x Treat	0.357***			0.115***			0.242***		
	(0.075)			(0.040)			(0.063)		
Articles x Treat		0.163***			0.031**			0.132***	
		(0.029)			(0.012)			(0.024)	
Google x Treat			0.091***			0.026***			0.065***
			(0.016)			(0.007)			(0.013)
N	535583	535583	535583	535583	535583	535583	535583	535583	535583

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the matched difference-in-difference estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effects. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. Note that the models here use a 1:5 matched sample for each type of crime. Standard errors are clustered at the level of the matched neighbourhoods.

Table 9: Impact of Muslim Salience on Crime for Domestic/Foreign Attacks

	(1) Tot	(2) Tot	(3) Viol	(4) Viol	(5) NViol	(6) NViol
Dom. Attack x Treat	0.073*	0.085**	0.082***	0.092***	-0.010	-0.008
	(0.039)	(0.039)	(0.021)	(0.021)	(0.030)	(0.030)
Int. Attack x Treat	0.077**	0.104***	0.009	0.018	0.068**	0.086***
	(0.036)	(0.036)	(0.020)	(0.020)	(0.029)	(0.029)
Dom. Neigh. x Treat		0.043***		0.036***		0.007
		(0.017)		(0.008)		(0.013)
Int. Neigh. x Treat		0.096***		0.032***		0.064***
		(0.015)		(0.008)		(0.012)
N	2745211	2745211	2745211	2745211	2745211	2745211

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of major Islamic terrorist attacks (either domestic or international) on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent, and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 10: Impact of Muslim Salience on Crime by Austerity Losses

	(1) Tot	(2) Tot	(3) Viol	(4) Viol	(5) NViol	(6) NViol
Attack x Treat	-0.146	0.160***	-0.125	0.070**	-0.020	0.090**
	(0.197)	(0.051)	(0.082)	(0.029)	(0.152)	(0.038)
Attack x Neigh.	0.084**	0.125***	0.038**	0.037**	0.045	0.088***
	(0.039)	(0.028)	(0.016)	(0.014)	(0.032)	(0.021)
N	520440	511404	520440	511404	520440	511404

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for per person loss from austerity policies at the local authority district level. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 11: Muslim Salience and Total Crime by Economics Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pop. Dens.	Pop. Dens.	Educ.	Educ.	LR Unemp.	LR Unemp.	Soc. Housing	Soc. Housing
Attack x Treat	0.336	0.086*	0.198	0.054	-0.439	0.110**	0.154	0.073
	(0.275)	(0.047)	(0.155)	(0.046)	(0.322)	(0.044)	(0.125)	(0.051)
Attack x Neigh.	0.106**	0.092***	0.056	0.087***	0.088***	0.091***	0.028	0.060**
	(0.043)	(0.027)	(0.035)	(0.025)	(0.032)	(0.026)	(0.034)	(0.029)
N	689983	685928	685720	687429	689986	684110	688643	684748

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for population density, education, long-term unemployment and social housing at the parliamentary constituency level. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 12: Muslim Salience and Violent Crime by Economics Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pop. Dens.	Pop. Dens.	Educ.	Educ.	LR Unemp.	LR Unemp.	Soc. Housing	Soc. Housing
Attack x Treat	0.305**	0.048*	0.190**	-0.005	0.002	0.046*	0.117*	0.033
	(0.130)	(0.025)	(0.083)	(0.023)	(0.098)	(0.025)	(0.062)	(0.028)
Attack x Neigh.	0.055**	0.032**	0.045**	0.021*	0.033*	0.037***	-0.003	0.009
	(0.026)	(0.014)	(0.018)	(0.012)	(0.018)	(0.014)	(0.014)	(0.014)
N	689983	685928	685720	687429	689986	684110	688643	684748

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for population density, education, long-term unemployment and social housing at the parliamentary constituency level. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 13: Muslim Salience and Non-violent Crime by Economics Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Pop. Dens.	Pop. Dens.	Educ.	Educ.	LR Unemp.	LR Unemp.	Soc. Housing	Soc. Housing
Attack x Treat	0.030	0.037	0.009	0.060*	-0.441*	0.063*	0.037	0.040
	(0.218)	(0.036)	(0.123)	(0.035)	(0.266)	(0.034)	(0.100)	(0.038)
Attack x Neigh.	0.051	0.060***	0.010	0.066***	0.054**	0.054***	0.030	0.052**
	(0.033)	(0.021)	(0.026)	(0.020)	(0.026)	(0.020)	(0.029)	(0.022)
N	689983	685928	685720	687429	689986	684110	688643	684748

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for population density, education, long-term unemployment and social housing at the parliamentary constituency level. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 14: Muslim Salience and Total Crime by Social Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Relig. Frac.	Relig. Frac.	Foreign	Foreign	D. Foreign	D. Foreign	Africa/ME Born	Africa/ME Born
Attack x Treat	0.568**	0.100**	0.023	0.520**	0.402**	0.095**	0.503*	0.065*
	(0.266)	(0.040)	(0.040)	(0.222)	(0.192)	(0.042)	(0.260)	(0.039)
Attack x Neigh.	-0.014	0.115***	0.058**	0.011	0.010	0.056**	0.027	0.079***
	(0.045)	(0.026)	(0.025)	(0.048)	(0.061)	(0.026)	(0.056)	(0.025)
N	690539	684736	684506	688090	690618	684427	688798	685131

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for religious fractionalization, share of non-UK born individuals, the change in the share of non-UK born individuals, and the share of residents born in Africa and the Middle East at the parliamentary constituency level. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 15: Muslim Salience and Violent Crime by Social Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Relig. Frac.	Relig. Frac.	Foreign	Foreign	D. Foreign	D. Foreign	Africa/ME Born	Africa/ME Born
Attack x Treat	0.314**	0.045**	-0.026	0.340***	0.269***	0.042*	0.269*	0.016
	(0.130)	(0.023)	(0.022)	(0.100)	(0.095)	(0.024)	(0.143)	(0.022)
Attack x Neigh.	-0.002	0.043***	0.002	0.022	0.059**	0.011	0.003	0.016
	(0.024)	(0.013)	(0.012)	(0.024)	(0.024)	(0.013)	(0.027)	(0.013)
N	690539	684736	684506	688090	690618	684427	688798	685131

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

*Notes:*This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for religious fractionalization, share of non-UK born individuals, the change in the share of non-UK born individuals, and the share of residents born in Africa and the Middle East at the parliamentary constituency level.

Table 16: Muslim Salience and Non-violent Crime by Social Factors

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Relig. Frac.	Relig. Frac.	Foreign	Foreign	D. Foreign	D. Foreign	Africa/ME Born	Africa/ME Born
Attack x Treat	0.254	0.055*	0.049	0.179	0.133	0.054*	0.235	0.049*
	(0.219)	(0.030)	(0.030)	(0.214)	(0.200)	(0.032)	(0.214)	(0.029)
Attack x Neigh.	-0.012	0.072***	0.056***	-0.012	-0.049	0.045**	0.024	0.063***
	(0.035)	(0.021)	(0.020)	(0.037)	(0.054)	(0.020)	(0.044)	(0.020)
N	690539	684736	684506	688090	690618	684427	688798	685131

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is in the bottom 25th or top 75th percentile for religious fractionalization, share of non-UK born individuals, the change in the share of non-UK born individuals, and the share of residents born in Africa and the Middle East at the parliamentary constituency level.

Table 17: Impact of Muslim Salience on Crime by Political Preferences (2010 GE)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Tot	Tot	Tot	Tot	Viol	Viol	Viol	Viol	NViol	NViol	NViol	NViol
Attack x Treat	0.111**	0.080	0.033	0.159***	0.039*	0.031	0.061	0.104***	0.072**	0.049	-0.028	0.055
	(0.046)	(0.083)	(0.120)	(0.057)	(0.023)	(0.048)	(0.064)	(0.031)	(0.034)	(0.059)	(0.111)	(0.045)
Attack x Neigh.	0.090***	0.014	0.023	0.088***	0.036***	0.015	0.042	0.049***	0.054***	-0.002	-0.018	0.039**
	(0.019)	(0.043)	(0.093)	(0.021)	(0.009)	(0.020)	(0.041)	(0.011)	(0.015)	(0.030)	(0.059)	(0.017)
N	1381065	1381065	1381065	1364146	1381065	1381065	1381065	1364146	1381065	1381065	1381065	1364146

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is above or below the median in terms of voting for right-wing parties in the 2010 general election (UKIP or BNP) at the parliamentary constituency level. Note that third and fourth model for each crime type are re-weighted first on economic and then on both economic and social variables. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 18: Impact of Muslim Salience on Crime by Political Preferences (2015 GE)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Tot	Tot	Tot	Tot	Viol	Viol	Viol	Viol	NViol	NViol	NViol	NViol
Attack x Treat	0.098**	0.120	0.030	0.213***	0.043*	0.094*	0.069	0.119***	0.055*	0.026	-0.039	0.094*
	(0.043)	(0.102)	(0.123)	(0.064)	(0.022)	(0.052)	(0.066)	(0.034)	(0.033)	(0.073)	(0.128)	(0.051)
Attack x Neigh.	0.108***	0.039	0.081	0.072***	0.046***	0.042*	0.040	0.039***	0.063***	-0.003	0.041	0.033*
	(0.018)	(0.042)	(0.108)	(0.023)	(0.009)	(0.022)	(0.049)	(0.012)	(0.014)	(0.030)	(0.074)	(0.018)
N	1390224	1390224	1390224	1354987	1390224	1390224	1390224	1354987	1390224	1390224	1390224	1354987

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is above or below the median in terms of voting for UKIP in the 2015 general election at the parliamentary constituency level. Note that third and fourth model for each crime type are re-weighted first on economic and then on both economic and social variables. Standard errors are reported in parentheses and are clustered at the neighbourhood-level for the first two models, and at the parliamentary constituency level afterwards for each crime type.

Table 19: Impact of Muslim Salience on Crime by Political Preferences (Brexit)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Tot	Tot	Tot	Tot	Viol	Viol	Viol	Viol	NViol	NViol	NViol	NViol
Attack x Treat	0.067	0.035	0.239*	0.234***	0.028	0.097**	0.118	0.128***	0.039	-0.063	0.121	0.106**
	(0.044)	(0.121)	(0.130)	(0.061)	(0.022)	(0.047)	(0.083)	(0.033)	(0.033)	(0.090)	(0.094)	(0.048)
Attack x Neigh.	0.092***	0.078*	0.186**	0.087***	0.037***	0.066*	0.081**	0.048***	0.055***	0.012	0.105*	0.039**
	(0.019)	(0.045)	(0.090)	(0.022)	(0.009)	(0.038)	(0.037)	(0.011)	(0.015)	(0.029)	(0.058)	(0.017)
N	1398508	1398508	1398508	1346703	1398508	1398508	1398508	1346703	1398508	1398508	1398508	1346703

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. The results are split by whether a neighbourhood is above or below the median in terms of voting for Brexit at the parliamentary constituency level. Note that third and fourth model for each crime type are re-weighted first on economic and then on both economic and social variables. Standard errors are reported in parentheses and are clustered at the neighbourhood-level for the first two models, and at the parliamentary constituency level afterwards for each crime type.

Table 20: Robustness Checks - Alternative Treatment Definitions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Tot	Tot	Tot	Viol	Viol	Viol	NViol	NViol	NViol
Attack x Q75 Mosq.	0.106***			0.055***			0.051*		
	(0.036)			(0.019)			(0.028)		
Attack x Q90		0.057**			0.052***			0.006	
		(0.023)			(0.012)			(0.018)	
Attack x Out.			0.076**			0.080***			-0.004
			(0.030)			(0.014)			(0.023)
N	2745211	2745211	2745211	2745211	2745211	2745211	2745211	2745211	2745211

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Three measure of exposure are used in these regressions: whether the size of the local Muslim community in the local authority is in the 75th percentile, and there is a Mosque in the vicinity (within 500m), whether the size of the local Muslim community in the local authority is in the 90th percentile, and whether the size of the Muslim community is an outlier in the local authority. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. Models (1)-(3) report the effects on the rate of total crime defined as the number of police reported offences per 1000 residents per month, models (4)-(6) focus on the rate of violent crime and models (7)-(9) concern the rate of non-violent crime. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 21: Robustness Checks - Placebo Exposure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Tot	Tot	Tot	Viol	Viol	Viol	NViol	NViol	NViol
Attack x Jewish	-0.023			-0.007			-0.017		
	(0.018)			(0.009)			(0.014)		
Attack x Sikh		-0.021			-0.006			-0.015	
		(0.017)			(0.008)			(0.013)	
Attack x Hindu			-0.035			0.008			-0.043**
			(0.023)			(0.010)			(0.019)
N	2745211	2745211	2745211	2745211	2745211	2745211	2745211	2745211	2745211

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. The measure of exposure used in these regression is whether the size of the local Jewish, Sikh or Hindu community in the local authority is in the 90th percentile. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. Models (1)-(3) report the effects on the rate of total crime defined as the number of police reported offences per 1000 residents per month, models (4)-(6) focus on the rate of violent crime and models (7)-(9) concern the rate of non-violent crime. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 22: Robustness Checks - Control Variable Interactions

	(1) Tot	(2) Viol	(3) NViol
Attack x Treat	0.153***	0.054**	0.099***
	(0.046)	(0.023)	(0.036)
Attack x Neigh.	0.064***	0.016*	0.048***
	(0.017)	(0.008)	(0.014)
N	2745211	2745211	2745211

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline estimation results illustrating the average treatment effect of changes to Muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively. Each model include interaction terms for control variables at the neighbourhood-level. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table 23: Robustness Checks - Placebo Shocks

	(1) Tot	(2) Viol	(3) NViol
Average Effects	0.005	0.001	0.004
	(0.004)	(0.001)	(0.003)
N	1000	1000	1000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table illustrates the average coefficients from regressions based on 1000 simulated terrorist attack series. The models report the effects on the rate of total, violent and non-violent crime defined as the number of police reported offences per 1000 residents per month respectively.